WRITTEN PROJECT 1
BAN403

Spring, 2026

Start: *05.03.26 09:00*

End: *13.03.26 12:00*

THE HOME EXAMINATION SHOULD BE SUBMITTED IN WISEFLOW

You can find information on how to submit your paper here:
https://www.nhh.no/en/for-students/examinations/home-exams-and-assignments/

Your candidate number will be announced on StudentWeb. The candidate number should be noted on all pages (not your name or student number). In case of group examinations, the candidate numbers of all group members should be noted.

Collaboration between individuals or groups on submission preparation, as well as exchange of self-produced materials between individuals or groups is prohibited. The answer paper must consist of individual's or the group's own assessments and analysis. All communication during the home exam is considered cheating. All submitted assignments are processed in Ouriginal, a plagiarism control system used by NHH. Use of artificial intelligence (AI) in this exam is determined by the course coordinator. Information on AI usage and guidelines can be found on the course's Canvas page.

SUPPLEMENTARY REGULATIONS FOR HOME EXAMINATIONS

You can find supplementary regulations under the headline “Regulations”

https://www.nhh.no/en/for-students/regulations/

Find more information under chapter 4.0 in the Supplementary provisions to the regulations for fulltime study programmes

Number of pages, including front page: *9*

Number of attachments: *1* (poisson params Quatar 2022.csv)

Project #1

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Teaching assistant: Youkai Wu
Due Date: Friday, March 13 at 12:00 pm (noon)

Instructions to Students:

- The main report must be a pdf file named group-Assig1-BAN403.pdf, all the attachments must come in a zip file and have the prefix att-, for example, "att-group-proj1-BAN403.zip".
- You must deliver all the simulation and calculation files that you used to obtain any numerical result. Failing to do so will render those numerical results invalid, and they will be assessed as missing.
- The main body of the report must be limited to a maximum of 10 pages, and it has to be self-contained; failing to comply will be penalized. Appendices may be used as support but should not be essential for the flow and understanding of your answers. Diagrams can be included in the appendix.
- All answers provided in the report must be documented and justified properly. Answers with single numbers without justification will be considered incomplete.
- For this project, the work must be entirely the result of the team credited in the report. Collaboration is not allowed among groups. If in doubt, consider the guidelines outlined in Table 1.
- Any use of generative AI must be acknowledged and explained. It is your responsibility to explain to what extent it was used as a support. Read the AI regulations published on canvas before making any use of AI.
- All discrete event simulation models must be delivered in JaamSim.

Activity	Status	Description
Conceptual Discussions	Allowed	Discussing general simulation concepts, probability distributions, or theoretical formulas from the textbook.
Software Troubleshooting	Allowed	Helping another group find a specific menu item or understand a general error message in JaamSim.
Sharing Model Files	Not Allowed	Sending, viewing, or copying JaamSim files, simulation calculation files, or spreadsheets between groups.
Joint Modeling	Not Allowed	Sitting together with another group to build the simulation logic or flowchart structure simultaneously.
Document Sharing	Not Allowed	Sharing drafts, outlines, or written text for the final main report.
Comparing Results	Not Allowed	Comparing numerical results, probabilities, expected points, or expected goal differences with another group to verify answers.

Table 1: Acceptable vs. Unacceptable Collaboration

1 Monte Carlo simulation: 2022 FIFA World Cup (40 points)

This problem asks you to run a Monte Carlo simulation of the group stage of the 2022 FIFA World Cup using the provided Competing Poisson goal-scoring model. For every team, an attacking strength parameter α, a defensive parameter β, and home-effect coefficients (γ, δ) are provided in the file poisson params Quatar 2022.csv. Keep in mind that not all teams in the file were in the world cup, but the extra ones were needed for fine tuning.

For any match with home team i and away team j, let X_{ij} be the home-team goals and Y_{ij} be the away-team goals. Conditional on team parameters, goals are assumed independent and Poisson distributed:

$$\Pr(X_{ij} = x) = \frac{e^{-\lambda_{ij}} \lambda_{ij}^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

$$\Pr(Y_{ij} = y) = \frac{e^{-\mu_{ij}} \mu_{ij}^y}{y!}, \qquad y = 0, 1, 2, \ldots$$

The joint scoreline probability is

$$\Pr(X_{ij} = n,\ Y_{ij} = m) = \Pr(X_{ij} = n) \Pr(Y_{ij} = m) = \frac{e^{-\lambda_{ij}} \lambda_{ij}^n}{n!} \cdot \frac{e^{-\mu_{ij}} \mu_{ij}^m}{m!}, \qquad n, m = 0, 1, 2, \ldots$$

The goal intensities λ_{ij} and μ_{ij} are computed depending on whether a home effect is applied. With a home effect:

$$\lambda_{ij} = \alpha_i \cdot \beta_j \cdot \gamma_i, \qquad \mu_{ij} = \alpha_j \cdot \beta_i \cdot \delta_i;$$

without a home effect (neutral venue):

$$\lambda_{ij} = \alpha_i \cdot \beta_j, \qquad \mu_{ij} = \alpha_j \cdot \beta_i.$$

Treat all group-stage matches as neutral except those played against Qatar.

In this assignment, you must restrict match scorelines to 0–5 goals per team. For each match, enumerate all 36 outcomes $(x, y) \in \{0, \ldots, 5\}^2$, compute their probabilities using the formula above, and then renormalize those probabilities so they sum to 1 over the 36 outcomes; sample the match result from this renormalized distribution.

For the group stage, you must follow the real World Cup rules to compute points and rank teams (points, goal difference, goals scored, and head-to-head criteria as needed) to determine the top two qualifiers in each group. Details are available at https://en.wikipedia.org/wiki/2022_FIFA_World_Cup in the group stage section.

Simulate the group stage only. Do not simulate the knockout bracket. Your outputs concern group ranking outcomes.

Run at least 10000 simulation iterations and report the group qualification simulation results:

1. For each group, report each team's probability of qualifying (finishing in the top two)
2. Report the probability of a team finishing first in the group.
3. You also need to report the expected points and expected goal difference per team.
4. Is this a reasonable model to predict the results? Explain why.

2 Discrete event simulation: Global Cargo Insurance (GCI) – The Freight Claim Process (60 points)

Global Cargo Insurance (GCI) is a specialized insurance firm that underwrites international shipping and freight operations. Its primary mission is to protect shipping companies and logistics providers against financial losses due to damaged or lost cargo during transit. Recently, GCI management launched a strategic initiative to evaluate their "Commercial Freight Claim Fulfillment Process." Customers have been complaining about severe delays in claim payouts, and management wants to ensure the internal operations are efficient, effective, and capable of handling future growth.

2.1 The Commercial Freight Claim Fulfillment Process

The analyst observe the process and summarized her observations as follows. The process is triggered when a client (e.g., a commercial shipping line) experiences a cargo loss and files a claim. The client submits a formal "Freight Claim Request" (FCR), which includes damage reports, photographic evidence, original shipping manifests, and financial valuation documents.

Once the client submits the documentation, the FCR undergoes the following sequence of activities:

1. **Intake and Requisite Review:** The FCR arrives at GCI's central intake desk. An administrative clerk spends **20 minutes** verifying the documents to ensure all administrative requisites are met (e.g., correct policy numbers, signatures, and inclusion of the shipping manifest). If the file is incomplete, it is returned immediately to the client; this happens with approximately 5% of the applications. If complete, the clerk logs the claim into the "ClaimFlow" IT system; this happens with approximately 95% of the applications.

2. **Digital Transfer:** The file is routed digitally to the Marine Investigation Department. Due to legacy system latency and batch-notification delays, this handoff takes **15 minutes**.

3. **Damage Investigation Study:** A specialized marine investigator pulls the FCR from the queue. The investigator spends **3 days (1,440 minutes)** analyzing the photographic evidence, weather reports, and port logs to determine the root cause of the damage and establish liability. There are currently three investigators available.

4. **Liability Decision:**
   - If the investigator determines the damage was caused by gross negligence on the part of the client (which is not covered), the claim is declined and the process ends (Historically, this happens to **10%** of claims).
   - If the claim is deemed valid (**90%** of the time), the file proceeds to the next step.

5. **Handoff to Finance:** The validated claim is transferred to the Actuarial & Finance Department

6. **Financial Assessment:** A senior claims adjuster reviews the financial valuation of the damaged goods to calculate the exact payout. This review takes **1 day (480 minutes)**. There are currently 2 adjusters available.

7. **The Rework Loop:** After the financial assessment, the adjuster frequently (about **25% of the time**) discovers that the "Currency Conversion Form" submitted by the client uses an outdated exchange rate.
   - *Return Transfer:* The adjuster halts the assessment and sends the file back to the Intake Clerk.
   - *Clerk Wait Time:* The returned FCR sits in the clerk's inbox for **1 day (480 minutes)** before being addressed.
   - *Correction:* The clerk must contact the shipping client to obtain the updated currency form. This takes **2 days (960 minutes)**.
   - *Re-submission:* The FCR file is manually updated and the corrected file sent back to the Actuarial & Finance Department to restart the Financial Assessment (**15 minutes**).

8. **Payout Approval Decision:** Once the financial assessment is complete and the data is verified, a final approval decision is made. Approximately **5%** of claims are rejected at this stage due to suspected financial fraud, terminating the process. The remaining **95%** are approved for payout.

9. **Settlement Communication:** For approved claims, the Legal Department drafts the formal "Deed of Release and Settlement" detailing the payout amount and terms. This takes about **1 day (480 minutes)**. There is currently one lawyer dedicated to this task.

10. **Handoff to Treasury (Transportation):** Upon completion, the settlement documents are routed to the Treasury Department. This takes **15 minutes**.

11. **Funds Transfer (Operation):** The Treasury Department processes the FCR and releases the funds to the client's bank account. This final activity takes **30 minutes**.

2.2 Simulation Data and Probability Distributions

For the Discrete Event Simulation, the deterministic average times must be replaced with the following probability distributions to accurately model the process dynamics and system variability.

2.2.1 The Arrival Process

- **Claim Arrivals:** The arrival of new Freight Claim Requests (FCRs) follows a Poisson process. The interarrival times between claims are **Exponentially distributed** with a mean of **120 minutes** (which equates to an average of 4 claims arriving per 8-hour workday).

2.2.2 Activity Processing Times

1. **Intake and Requisite Review:** The time to manually review the documents varies slightly depending on the clerk's focus and the file's legibility. Assume this time is **Normally distributed** with a mean of **20 minutes** and a standard deviation of **4 minutes**.

2. **Digital Transfer:** System latency and batch-notification delays are not perfectly constant. Model this using a **Gamma distribution** with a shape parameter (k or α) of 5 and a scale parameter (θ or β) of 3. (Mean = **15 minutes**).

3. **Damage Investigation Study:** The complexity of marine investigations varies wildly, but it always requires a minimum baseline of effort before stretching into a long right-tail of complex cases. Model this using a **Gamma distribution** with a shape parameter (α) of 2 and a scale parameter (β) of 720 minutes. (Mean = **1,440 minutes** or 3 days).

4. **Gateway 1 (Liability Decision): Discrete Probability:** 10% decline, 90% proceed.

5. **Handoff to Finance: Constant (Deterministic)** time of exactly **15 minutes**.

6. **Financial Assessment:** Adjusters have a highly standardized method for calculating valuations, but the time is bounded by strict minimum and maximum limits based on the number of items on a manifest. Model this using a scaled **Beta distribution** bounded between a minimum of **360 minutes** and a maximum of **600 minutes**, using shape parameters $\alpha = 2$ and $\beta = 2$ to create a symmetric curve. (Mean = **480 minutes**).

7. **The Rework Loop (Triggered 25% of the time):**

   - **Return Transfer: Constant** 15 minutes.
   - **Clerk Wait Time (Inbox Delay):** The time the file sits before the clerk checks the inbox and begins the correction typically occurs in phases or batches. Model this delay using an **Erlang distribution** with a shape parameter (k) of 3 and a scale parameter (θ) of 160 minutes. (Mean = **480 minutes** or 1 day).
   - **Correction:** Contacting the client and updating the currency forms takes time. Assume this is **Normally distributed** with a mean of **2 days (960 minutes)** and a standard deviation of **120 minutes**.
   - **Re-submission: Constant** 15 minutes.

8. **Gateway 2 (Payout Approval): Discrete Probability:** 5% decline, 95% approved.

9. **Settlement Communication:** Legal drafting is relatively standard but requires some customization. Model this as **Uniformly distributed** between **420 and 540 minutes** (Mean = 480 minutes).

10. **Handoff to Treasury: Constant (Deterministic)** time of exactly **15 minutes**.

11. **Funds Transfer:** The final transaction is quick but depends on banking network speeds. Assume this is **Normally distributed** with a mean of **30 minutes** and a standard deviation of **5 minutes**.

2.3 The Business Challenge

GCI's Chief Operating Officer (COO) is convinced that the staffing numbers are correct and suspects that the current workflow suffers from unnecessary functional silos, a lack of process ownership, and costly rework loops that use up expensive resources (like the Senior Adjusters). The COO has hired your consulting team to analyze this process. Using the process mapping, analytical capacity models, and discrete event simulation concepts, your team must:

1. Document the "AS-IS" process architecture. Translate the messy real world into a clear process architecture. Your tasks are:

   - Define boundaries.
   - Identify the flow unit.

- Identify resources (people, equipment, IT systems) and buffers (queues, waiting rooms, inboxes). Use the following required tools: general process chart and service system map (SSM) with customer/front-office/back-office bands.
- Classify the value-adding, non-value-adding, and business value adding activities.
- Write a short case for action: what is wrong with the current process? (e.g., long waits, errors, poor utilization, complaints). What business or operational impact does that have?
- Write a vision statement for the improved process (2–3 sentences).
- Define 3 quantitative performance measures (KPIs), linked to the "devil's quadrangle": time, cost, quality, flexibility (e.g., average waiting time, resource utilization, percentage of customers served within target time).
- **Deliverable** 1–2 page problem definition & objectives section. The charts can be in the appendix but their must be explicitly use as support for your description in the main document body.

2. Detailed workflow documentation: Create a more detailed flowchart diagram. Also, organize all the data provided in the description and explain how it will be used. **Deliverable**: Detailed flowchart.

3. Baseline analytical analysis

   - Identify bottleneck resource(s) by comparing demand vs capacity, as suggested in the workflow & line-balancing discussion in chapter4.
   - Summarize key baseline insights: where are the big delays, how variable is demand/processing, where is capacity under- or over-used?
   - **Deliverable**: Short "Baseline analysis" section.

4. Propose a "TO-BE" design that utilizes parallel processing and mistake-proofing (to capture currency errors at the source).

   - Use people-oriented & conceptual design principles from chapter3
     - Organize work around outcomes, not isolated tasks.
     - Reduce handoffs and sequential processing (case management, parallelization).
     - Capture information once at the source (eliminate re-entry).
     - Design for the dominant flow, not exceptions; mistake-proof critical steps.
   - Combine this with workflow design principles in chapter4 (product orientation, elimination of buffers, minimizing multiple paths and handoffs).
   - **Deliverable**: 1–2 conceptual redesigned process diagrams + short justification using named principles.

5. Build a simulation model of the current process description and your suggested improvements to demonstrate how your new design will perform dynamically under variable arrival rates and service times.

(a) Define simulation entities, attributes, activities, events, and state variables following the guidelines in chapter7

- Entities: customers, orders, patients, etc.
- Attributes: arrival time, service type, priority, etc.
- State variables: queue lengths, server status, number in system.

(b) Implement:

- arrival process with appropriate distribution(s)
- service time distributions at each station
- routing logic, balking/reneging if relevant.

(c) For the as-is design, verify and validate:

- Check that the logic matches the flowchart.
- Ensure basic output (utilization, average waiting times) is plausible and consistent with your earlier analysis.

(d) **Deliverable**: Simulation model files in JaamSim (the as-is and proposed improvements must be added as separate models) plus 1–2 pages describing structure, key assumptions, and validation checks.